

1 Summary

Model (Run-name): model1 (test)

Iteration: 22

TAC: 154867.23461911708

Took: 2.6726558208465576s

Total Time: 34.663506746292114s

Epsilons:

- ε_{LMTD} : 1e-06
- $\varepsilon_{Balancing}$: 1e-06
- ε_{Area} : 1e-06
- ε_{Beta} : 1e-06

Gurobi Tolerances:

- IntFeasTol: 1e-05
- FeasibilityTol: 1e-06
- OptimalityTol: 1e-06

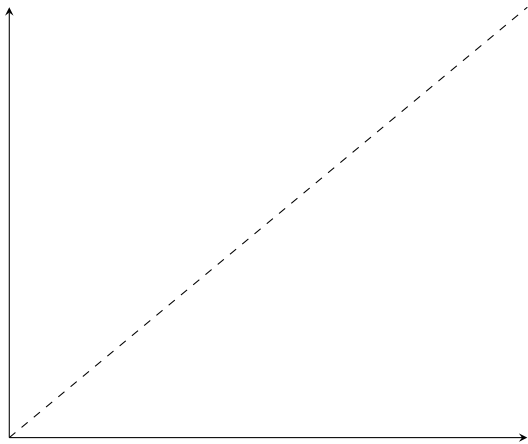
Gamsworld Solution: http://www.gamsworld.org/minlp/minlplib2/html/heatexch_gen1.html

2 Results

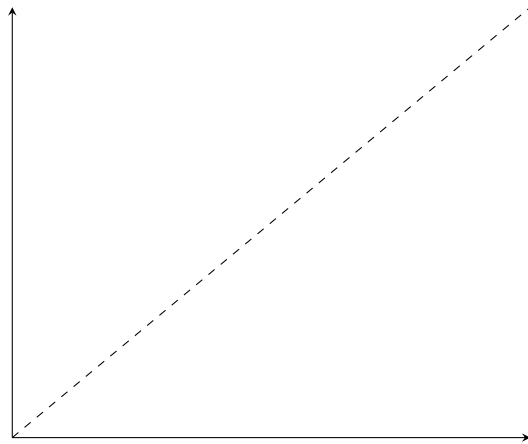
Variable	Value	Variable	Value	Variable	Value
$z_{1,1,1}$	1.0	$RecLMTD_{1,1,1}$	0.05225388243084932	$b_{1,1,1}^{H,out}$	897.3756331309916
$z_{1,2,2}$	1.0	$RecLMTD_{1,2,2}$	0.017654567489586874	$b_{1,2,2}^{H,out}$	3401.6451958696166
$z_{2,1,2}$	1.0	$RecLMTD_{2,1,2}$	0.02918142056761671b	$b_{2,1,2}^{H,out}$	4089.101064022653
$z_{cu,1}$	1.0	$RecLMTD_{cu,1}$	0.014591072513466603	$b_{1,1,1}^{C,in}$	8581.35738941887
$z_{cu,2}$	1.0	$RecLMTD_{cu,2}$	0.009582847254299868	$b_{1,2,2}^{C,in}$	4550.0
$z_{hu,1}$	1.0	$RecLMTD_{hu,1}$	0.02255517520436296	$b_{2,1,2}^{C,in}$	5863.861939186698
$A_{1,1,1}^\beta$	70.83181816751056	$q_{1,1,1}$	679.0950737207504	$b_{1,1,1}^{C,out}$	9260.45246313962
$A_{1,2,2}^\beta$	68.85281320938876	$q_{1,2,2}$	1950.0	$b_{1,2,2}^{C,out}$	6500.0
$A_{2,1,2}^\beta$	141.79783607898793	$q_{2,1,2}$	2431.357389418875	$b_{2,1,2}^{C,out}$	8295.219328605574
$A_{cu,1}^\beta$	4.962276317638335	$q_{cu,1}$	170.9049262792496		
$A_{cu,2}^\beta$	37.71812813120871	$q_{cu,2}$	1968.6426105811252		
$A_{hu,1}^\beta$	13.238585544475612	$q_{hu,1}$	489.54753686037475		
$A_{1,1,1}$	70.83181816751056	$t_{1,1}^{(H)}$	650.0		
$A_{1,2,2}$	68.85281320938876	$t_{1,2}^{(H)}$	582.090492627925		
$A_{2,1,2}$	141.79783607898793	$t_{1,3}^{(H)}$	387.09049262792496		
$A_{cu,1}$	4.962276317638335	$t_{2,1}^{(H)}$	590.0		
$A_{cu,2}$	37.71812813120871	$t_{2,2}^{(H)}$	590.0		
$A_{hu,1}$	13.238585544475612	$t_{2,3}^{(H)}$	468.4321305290563		
$dt_{1,1,1}$	32.63650245735834	$t_{1,1}^{(C)}$	617.3634975426417		
$dt_{1,1,2}$	10.0	$t_{1,2}^{(C)}$	572.090492627925		
$dt_{1,2,2}$	82.09049262792496	$t_{1,3}^{(C)}$	410.0		
$dt_{1,2,3}$	37.09049262792496	$t_{2,1}^{(C)}$	500.0		
$dt_{2,1,2}$	17.90950737207504	$t_{2,2}^{(C)}$	500.0		
$dt_{2,1,3}$	58.43213052905625	$t_{2,3}^{(C)}$	350.0		
$dt_{cu,1}$	67.09049262792496	$t_{1,1,1}^H$	370.0		
$dt_{cu,2}$	148.43213052905625	$t_{1,2,2}^H$	370.0		
$dt_{hu,1}$	62.63650245735832	$t_{2,1,2}^H$	370.0		
$f_{1,1,1}^H$	2.42533954900268	$t_{1,1,1}^C$	617.3634975426419		
$f_{1,2,2}^H$	9.193635664512477	$t_{1,2,2}^C$	500.0		
$f_{2,1,2}^H$	11.051624497358523	$t_{2,1,2}^C$	581.2340311616574		
$f_{1,1,1}^C$	15.0	$b_{1,1,1}^{H,in}$	1576.4707068517419		
$f_{1,2,2}^C$	13.0	$b_{1,2,2}^{H,in}$	5351.645195869616		
$f_{2,1,2}^C$	14.302102290699263	$b_{2,1,2}^{H,in}$	6520.458453441528		

3 *RecLMTD* Tangent Points

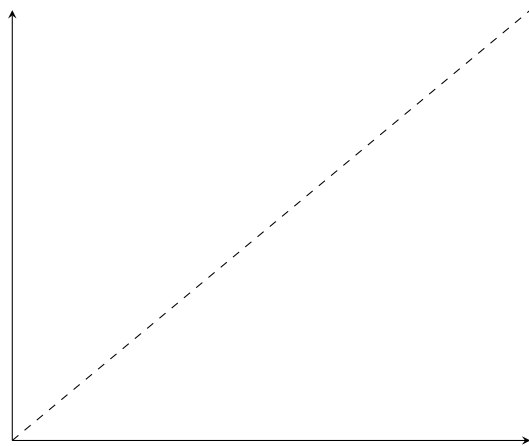
- $(1, 1, 1)$
 - Value: 0.052254
 - Error: 0.000000050281728
 - Relative error: 0.000000962257300
- $(1, 2, 2)$
 - Value: 0.017655
 - Error: 0.000000133266839
 - Relative error: 0.000007548518735
- $(2, 1, 2)$
 - Value: 0.029181
 - Error: 0.000000653956630
 - Relative error: 0.000022409531918
- $cu, 1$
 - Value: 0.014591
 - Error: 0.000000022225399
 - Relative error: 0.000001523216722
- $cu, 2$
 - Value: 0.009583
 - Error: 0.000000375503737
 - Relative error: 0.000039183450788
- $hu, 1$
 - Value: 0.022555
 - Error: 0.000000882504727
 - Relative error: 0.000039124954293



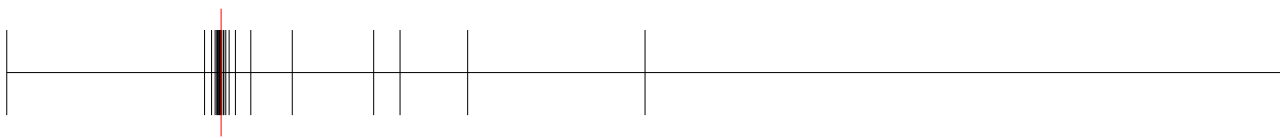
(a) *RecLMTD* - (1, 1, 1)



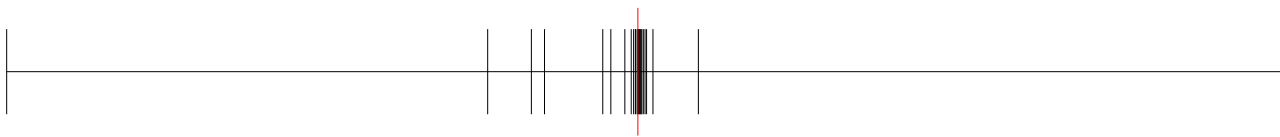
(b) *RecLMTD* - (1, 2, 2)



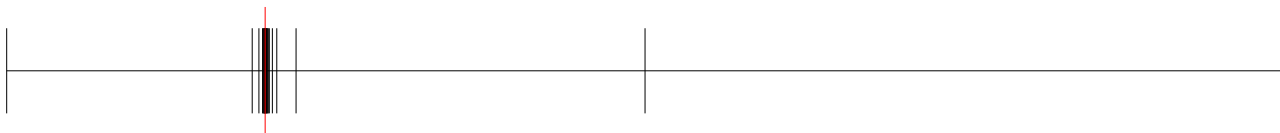
(c) *RecLMTD* - (2, 1, 2)



(a) *RecLMTD* - cu, 1



(b) *RecLMTD* - cu, 2



(a) *RecLMTD* - hu, 1

4 Balancing Breakpoints

4.1 $b^{H,\text{in}}$

- (1, 1, 1)
 - Value: 1576.4707068517419
 - Absolute error: 0.0
 - Relative error: 0.0
- (1, 2, 2)
 - Value: 5351.645195869616
 - Absolute error: 0.11728287188816466
 - Relative error: 2.1915773176350143e-05
- (2, 1, 2)
 - Value: 6520.458453441528
 - Absolute error: 0.0
 - Relative error: 0.0

4.2 $b^{H,\text{out}}$

- (1, 1, 1)
 - Value: 897.3756331309916
 - Absolute error: 0.0
 - Relative error: 0.0
- (1, 2, 2)
 - Value: 3401.6451958696166
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 1, 2)
 - Value: 4089.101064022653
 - Absolute error: 0.0
 - Relative error: 0.0

4.3 $b^{C,\text{in}}$

- (1, 1, 1)
 - Value: 8581.35738941887
 - Absolute error: 3.637978807091713e-12
 - Relative error: -4.2393978504816423e-16
- (1, 2, 2)
 - Value: 4550.0
 - Absolute error: 0.0

- Relative error: 0.0
- (2, 1, 2)
 - Value: 5863.861939186698
 - Absolute error: 0.0
 - Relative error: 0.0

4.4 $b^{C,out}$

- (1, 1, 1)
 - Value: 9260.45246313962
 - Absolute error: 9.094947017729282e-12
 - Relative error: -9.82127715025899e-16
- (1, 2, 2)
 - Value: 6500.0
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 1, 2)
 - Value: 8295.219328605574
 - Absolute error: 17.649239903934358
 - Relative error: -0.002123122693265299



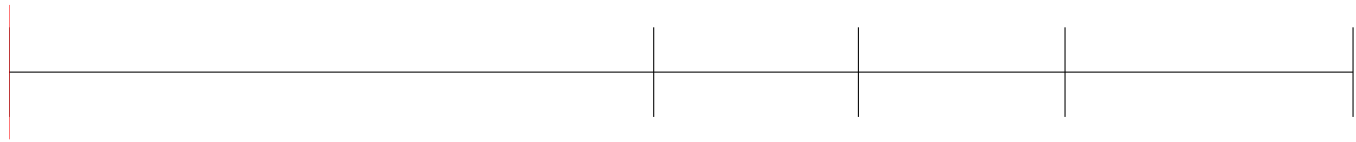
(a) $t^{(H)} - (1, 1)$



(a) $t^{(H)} - (1, 2)$



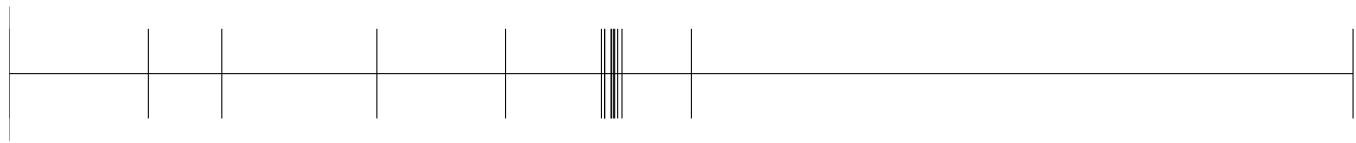
(a) $t^{(H)} - (2, 2)$



(a) $t^{(H)} - (1, 1, 1)$



(a) $t^{(H)} - (1, 2, 2)$



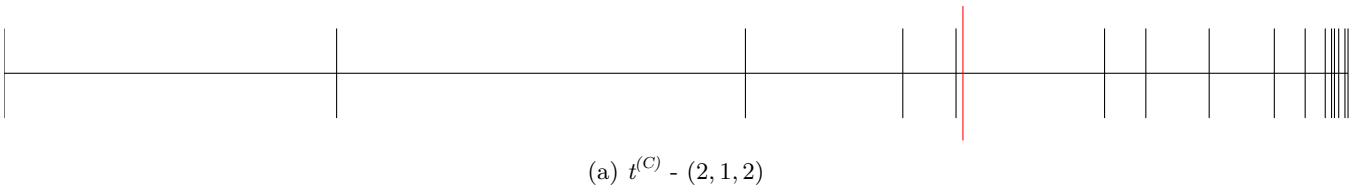
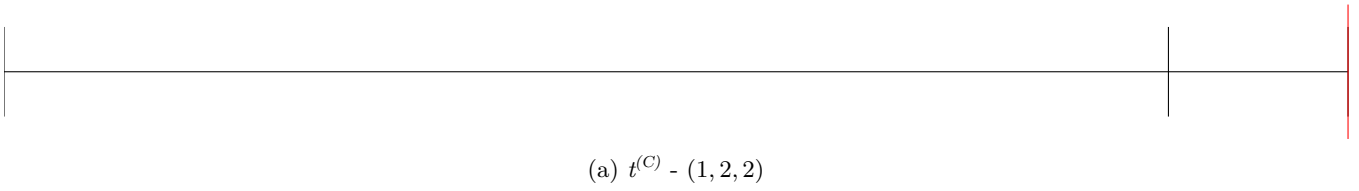
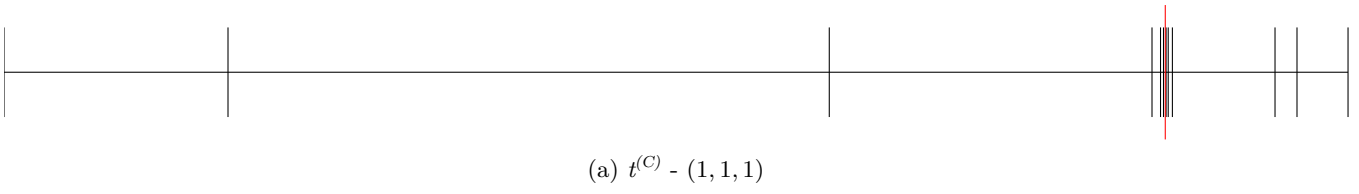
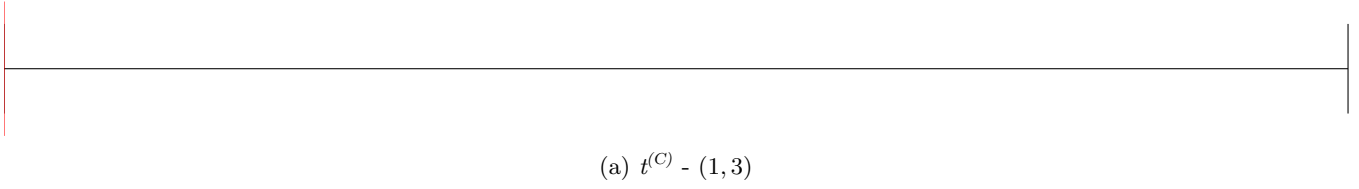
(a) $t^{(H)} - (2, 1, 2)$



(a) $t^{(C)} - (1, 2)$



(a) $t^{(C)} - (2, 3)$



5 Area Breakpoints

5.1 A

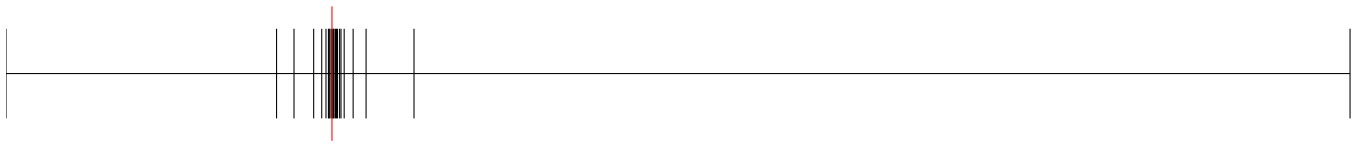
- $(1, 1, 1)$
 - Value: 70.831818
 - Error: 0.138890115635533
 - Relative error: 0.001957006193054
- $(1, 2, 2)$
 - Value: 68.852813
 - Error: 0.000000000000057
 - Relative error: 0.000000000000001
- $(2, 1, 2)$
 - Value: 141.797836
 - Error: 0.103088982641736
 - Relative error: 0.000726485627891

5.2 A_{CU}

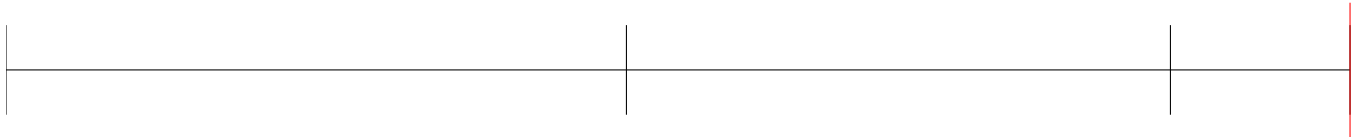
- $cu, 1$
 - Value: 4.962276
 - Error: 0.025096026860054
 - Relative error: 0.005031913626368
- $cu, 2$
 - Value: 37.718128
 - Error: 0.012274739801413
 - Relative error: 0.000325327557285

5.3 A_{HU}

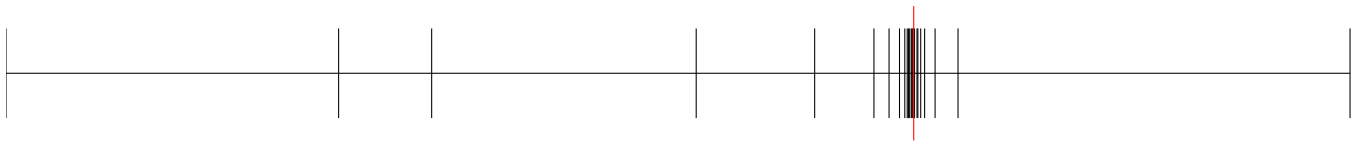
- $hu, 1$
 - Value: 13.238586
 - Error: 0.011611013224492
 - Relative error: 0.000876289885507



(a) $A^\beta - (1, 1, 1)$



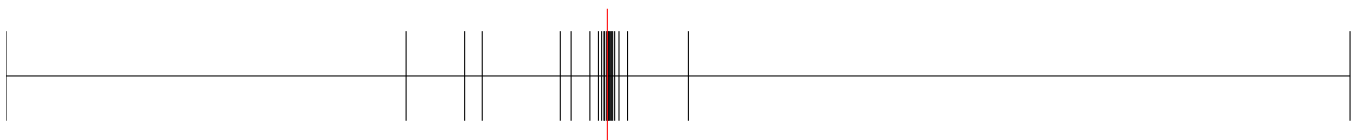
(a) $A^\beta - (1, 2, 2)$



(a) $A^\beta - (2, 1, 2)$



(a) $A^\beta - cu, 1$



(a) $A^\beta - cu, 2$

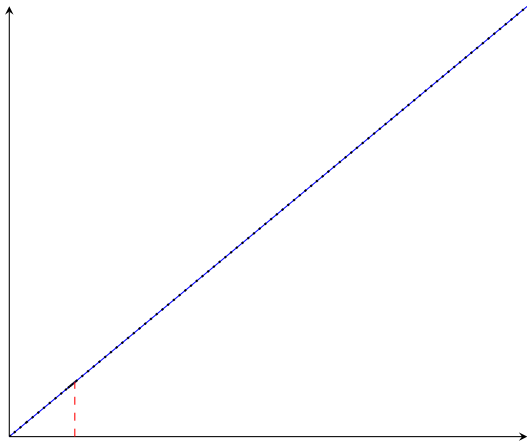


(a) $A^\beta - hu, 1$

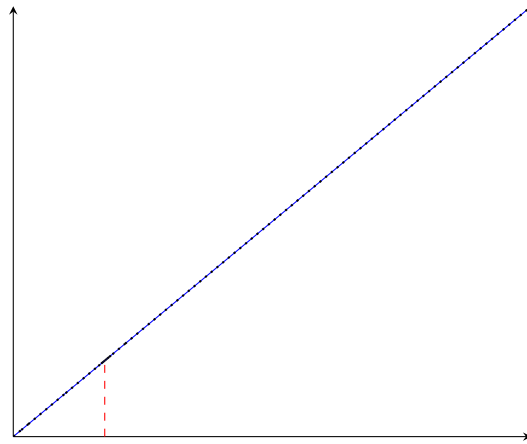
6 A^β Breakpoints

- (1,1,1)
 - q : 70.831818
 - $\hat{q}^\beta$: 70.831818
 - q^β : 70.831818
 - Breakpoints: [0, 5.0, 18.13353447520507, 60.38493803732795, 63.3468696286352, 65.40595072874316, 67.27362048222933, 69.14107665413653, 69.42777867315647, 69.89902342903967, 70.22290217272871, 70.30181758862284, 70.46247894104941, 70.50425945122109, **70.60541450524272**, **70.95870595008942**, 71.00935653192329, 71.22933662920877, 71.2616536557244, 71.33750386361461, 71.6637831223933, 71.88123217857378, 560.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- (1,2,2)
 - q : 68.852813
 - $\hat{q}^\beta$: 68.852813
 - q^β : 68.852813
 - Breakpoints: [0, 6.000977230027827, 11.262728503004022, 39.36524453694061, 64.01825678947372, 66.17325582523148, 67.27673099173637, 67.91344539650363, 68.10969007738701, **68.66676327863097**, **69.0314457108091**, 69.22655548799992, 69.34502840170047, 69.41355730879985, 70.08419437796795, 70.37078617087542, 70.94081287188642, 72.25348351595485, 74.6496383881616, 84.1468297226002, 390.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- (2,1,2)
 - q : 141.797836
 - $\hat{q}^\beta$: 141.797836
 - q^β : 141.797836
 - Breakpoints: [0, 9.898970403313276, 28.333333333333332, 42.69900440382105, 48.67452649027746, 79.83702775273511, 87.16107999267206, 118.88765652962957, 128.52021081477858, 131.69760457629607, 132.12228195956155, 135.81963163113545, 137.02923097052596, 139.47623233772285, 140.31685261310218, **140.7008934199032**, **142.4338700260237**, 144.67005099128556, 144.70843093515643, 147.07009017600456, 147.48374761404892, 149.84689443310864, 720.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- $cu, 1$
 - q : 4.962276
 - $\hat{q}^\beta$: 4.962276
 - q^β : 4.962276
 - Breakpoints: [0, 2.5450346292088537, 2.9819181007907165, 3.1809419553840232, 3.9948823402911255, 4.269280092716329, 4.432280503220532, 4.613047202566896, 4.707108999269609, 4.825601156120121, 4.849719301830286, **4.888121952381175**, **4.974201802143407**, 4.983938626310074, 4.993068227518925, 5.0418124100090695, 5.081618324442152, 7.702824927089129, 9.28003023700696, 13.56934656224007, 14.535545040173215, 181.61828057849593]
 - Error: 0.0000000000000000

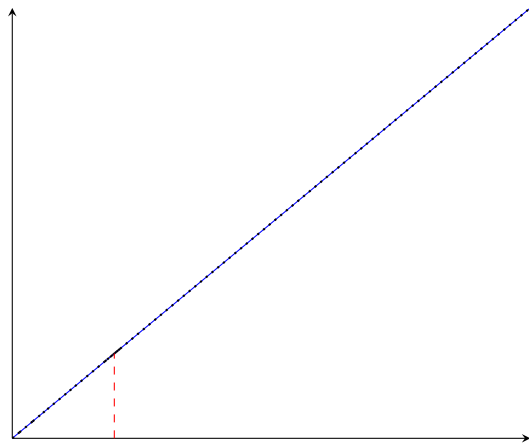
- Relative error: 0.0000000000000000
- $cu, 2$
 - q : 37.718128
 - $\hat{q}^\beta$: 37.718128
 - q^β : 37.718128
 - Breakpoints: [0, 18.023935870161154, 21.04660840930274, 24.97364426355679, 30.189424467922265, 34.23465128608717, 36.670233878040335, 37.21616748032837, 37.443498367229196, 37.47875108132638, 37.554741031910254, 37.612770977560544, 37.6551299822246, **37.682669717644174**, **37.76995088018704**, 37.78573239590169, 37.88148462263016, 37.9018730666985, 37.9454585315849, 38.10872593446926, 38.21652191748154, 38.88483572219945, 285.4001551947793]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- $hu, 1$
 - q : 13.238586
 - $\hat{q}^\beta$: 13.238586
 - q^β : 13.238586
 - Breakpoints: [0, 4.943755299006499, 12.544341108707222, 12.595096067947068, 12.917705220652834, 13.104034300095861, 13.161020122653593, 13.189852638760229, 13.192252946941991, 13.213749911872785, 13.218222920629291, **13.22717920580809**, **13.244985069124171**, 13.266498143734903, 13.269097009439808, 13.30871858348944, 13.32330648750218, 13.32663103908035, 13.346967780023082, 13.439500729482463, 13.468285806082731, 14.09178614751222, 237.3002543523117]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000



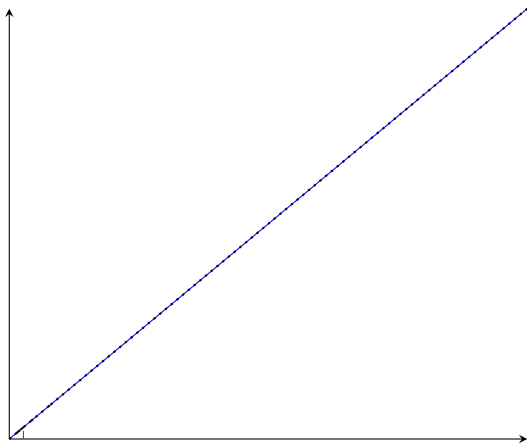
(a) $q^\beta - (1, 1, 1)$



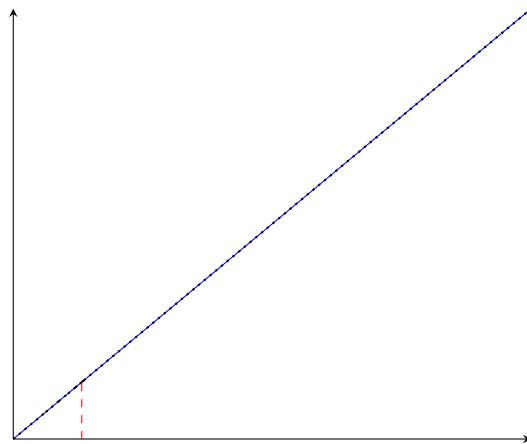
(b) $q^\beta - (1, 2, 2)$



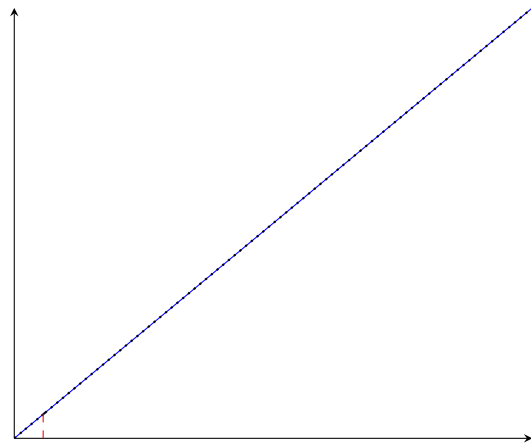
(c) $q^\beta - (2, 1, 2)$



(a) $q^\beta - cu, 1$



(b) $q^\beta - cu, 2$



7 Other variables

Variable	Value	Variable	Value	Variable	Value
$z_{1,1,2}$	-0.0	$f_{2,2,1}^C$	0.0	$b_{2,2,2}^{H,out}$	5279.541546558472
$z_{1,2,1}$	-0.0	$f_{2,2,2}^C$	0.0	$b_{1,1,2}^{C,in}$	286.138060813302
$z_{2,1,1}$	-0.0	$RecLMTD_{1,1,2}$	0.1	$b_{1,2,1}^{C,in}$	6500.0
$z_{2,2,1}$	-0.0	$RecLMTD_{1,2,1}$	0.008073038654985954	$b_{2,1,1}^{C,in}$	0.0
$z_{2,2,2}$	-0.0	$RecLMTD_{2,1,1}$	0.029166666666666688	$b_{2,2,1}^{C,in}$	0.0
$z_{hu,2}$	-0.0	$RecLMTD_{2,2,1}$	0.020111728838038183	$b_{2,2,2}^{C,in}$	0.0
$A_{1,1,2}^\beta$	0.0	$RecLMTD_{2,2,2}$	0.005479452054794521	$b_{1,1,2}^{C,out}$	286.1380608133004
$A_{1,2,1}^\beta$	0.0	$RecLMTD_{hu,2}$	0.005549329826813255	$b_{1,2,1}^{C,out}$	6500.0
$A_{2,1,1}^\beta$	0.0	$q_{1,1,2}$	0.0	$b_{2,1,1}^{C,out}$	0.0
$A_{2,2,1}^\beta$	0.0	$q_{1,2,1}$	0.0	$b_{2,2,1}^{C,out}$	0.0
$A_{2,2,2}^\beta$	0.0	$q_{2,1,1}$	0.0	$b_{2,2,2}^{C,out}$	0.0
$A_{hu,2}^\beta$	0.0	$q_{2,2,1}$	0.0		
$A_{1,1,2}$	0.0	$q_{2,2,2}$	0.0		
$A_{1,2,1}$	0.0	$q_{hu,2}$	0.0		
$A_{2,1,1}$	0.0	$t_{1,1,2}^H$	562.9371546047238		
$A_{2,2,1}$	0.0	$t_{1,2,1}^H$	650.0		
$A_{2,2,2}$	0.0	$t_{2,1,1}^H$	590.0		
$A_{hu,2}$	0.0	$t_{2,2,1}^H$	590.0		
$dt_{1,1,3}$	240.0	$t_{2,2,2}^H$	576.8824300983304		
$dt_{1,2,1}$	150.0	$t_{1,1,2}^C$	410.0		
$dt_{2,1,1}$	24.166666666666668	$t_{1,2,1}^C$	500.0		
$dt_{2,2,1}$	90.0	$t_{2,1,1}^C$	412.2735916996774		
$dt_{2,2,2}$	88.76712328767127	$t_{2,2,1}^C$	350.0		
$dt_{2,2,3}$	240.0	$t_{2,2,2}^C$	440.01120358475225		
$dt_{hu,2}$	180.0	$b_{1,1,2}^{H,in}$	469.2597304096333		
$f_{1,1,2}^H$	0.8063643354875234	$b_{1,2,1}^{H,in}$	4923.529293148258		
$f_{1,2,1}^H$	7.57466045099732	$b_{2,1,1}^{H,in}$	11800.0		
$f_{2,1,1}^H$	20.0	$b_{2,2,1}^{H,in}$	0.0		
$f_{2,2,1}^H$	0.0	$b_{2,2,2}^{H,in}$	5279.541546558472		
$f_{2,2,2}^H$	8.948375502641477	$b_{1,1,2}^{H,out}$	469.2597304096333		
$f_{1,1,2}^C$	0.6978977093007366	$b_{1,2,1}^{H,out}$	4923.529293148258		
$f_{1,2,1}^C$	13.0	$b_{2,1,1}^{H,out}$	11800.0		
$f_{2,1,1}^C$	0.0	$b_{2,2,1}^{H,out}$	0.0		